8.5.3 Methods of arc extinction

There are two methods of extinguishing the arc in circuit breakers.

1. High resistance method.
2. Low resistance method.

8.5.4 Classification of circuit breakers

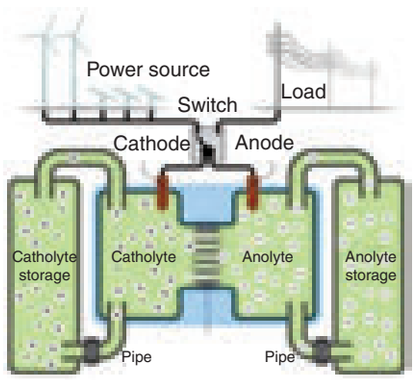
Circuit breakers are classified depending upon the medium use for arc extinction.

Sl. No.	Type	Medium used for arc extinction	Voltage range
1	Air break circuit breaker.	Air	Up to 1000 V
2	Tank type oil circuit breaker.	Transformer oil	Up to 33 KV
3.	Minimum oil circuit breaker.	Transformer oil	Up to 132 KV
4.	Air blast circuit breaker.	Compressed air	Up to 132 KV
5	Sf_6 circuit breaker.	Sf_6 gas	400 KV to 760KV
6	Vacuum circuit breaker.	Vacuum	11 KV to 33 KV
7	High voltage direct current circuit breaker.	sf_6 gas or vacuum	+ 500 KV DC

8.5.5 Difference between electrical isolator and electrical circuit breaker

Sl. No.	Electrical isolator	Electrical circuit breaker
1	Used only in no-load condition.	Used in either no-load or on-load condition.
2	Manual operation.	Automatic operation.
3	Mechanical equipment.	Electro-mechanical equipment.
4.	It break outs a portion of a substation when a fault occurred.	It trips the entire system, if any fault occurs.

Do you Know?



A redox flow battery uses excess solar- and wind-based energy to charge solutions of chemicals that can subsequently be stored for use when sunshine and wind are scarce. At that time, the chemical solutions of opposite charge can be pumped across solid electrodes, thus creating an electron exchange that provides power to the electrical grid.

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